

Volatility Forecasting for Risk-Aware Position Sizing

A Machine Learning Approach to Forward Volatility Estimation for Dynamic Position Management

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Machine Learning for Quantitative Investing Using Python

Generated: July 10, 2026

Abstract

This project documents a supervised learning approach to forecasting 5-day forward realized volatility for individual equities. The forecast drives a dynamic position-sizing rule where position scale is inversely proportional to predicted volatility. A ladder of three progressively sophisticated baselines—rolling historical volatility, exponentially weighted moving average (EWMA), and ridge-regularized linear regression—is established before evaluating LightGBM. A targeted experiment on leverage effect features improved calibration by 37.8%. Results indicate that LightGBM trained on a curated set of five features achieves the lowest out-of-sample RMSE (0.119) and is statistically unbiased under the Mincer-Zarnowitz framework ($p = 0.141$). A historical backtest demonstrates that dynamic sizing improves total return by 11.0 percentage points and reduces maximum drawdown by 11.3 percentage points compared to static sizing.

1. Introduction

Investment returns depend not only on predicting price direction but also on how much capital is allocated to each position. A systematic portfolio manager with a correct directional view can still lose money if a position is oversized during a period of unexpectedly high volatility. Conversely, a conservative position during an unexpectedly calm market wastes a high-conviction view. Many systematic strategies use static position sizing or reactive rules that adjust slowly to changing conditions, resulting in uneven risk contribution over time.

This project addresses the core problem of volatility forecasting for risk management. The objective is to produce a statistically evaluated, well-calibrated forecast of 5-day forward realized volatility for individual equities. The forecast is then used to drive a dynamic position-sizing rule: position scale equals target volatility divided by predicted volatility. The goal is not to predict price direction, but to quantify uncertainty about that direction.

The project follows a baseline-first methodology. Three progressively sophisticated baselines are established before any advanced model is considered: rolling historical volatility, exponentially weighted moving average (EWMA) volatility, and ridge-regularized linear regression. Complexity is justified only when it demonstrably outperforms simpler alternatives on pre-specified criteria. The advanced approach considered is LightGBM, a gradient-boosted tree model, evaluated alongside isolated feature engineering experiments.

2. Problem Formulation

The prediction target is 5-day forward annualized realized volatility. For each trading day t , the target is computed as the standard deviation of daily log returns over the subsequent 5 trading days, scaled by $\sqrt{252}$ to annualize the figure.

The economic context is a target volatility mandate of 15% annualized. Position sizing is the primary tool available to meet this mandate. The dynamic sizing rule is: $\text{position_scale} = \text{target_vol} / \text{predicted_vol}$. The scale is capped at a maximum leverage of 1.0 and floored at 0.0.

The project has four measurable objectives: (1) achieve lower out-of-sample RMSE and MAE than all three baselines; (2) produce forecasts that are statistically unbiased under the Mincer-Zarnowitz regression; (3) demonstrate that dynamic sizing improves Sharpe ratio and reduces maximum drawdown relative to static sizing; and (4) confirm forecast calibration using a volatility cone visualization.

3. Data and Feature Engineering

Data for 10 large-cap US equities (AAPL, MSFT, GOOGL, AMZN, META, JPM, XOM, JNJ, WMT, TSLA) and the $\wedge$ VIX index were sourced via yfinance from January 2020 through December 2024, yielding approximately 1,257 trading days. Daily OHLCV prices were used for feature construction.

Features were grouped into four categories: historical realized volatility (rolling windows of 21, 63, and 252 days), EWMA volatility ($\lambda=0.94$), range-based estimators (Parkinson volatility), and market context (VIX level and 5-day change). Advanced features included volatility regime, VIX-times-rolling-vol interaction, leverage effect (negative returns amplified), volatility of volatility, return reversal, and sector-relative volatility.

Correlation analysis was performed to identify redundant features. Features with absolute correlation greater than 0.8 were removed. The optimized feature sets were: Baseline 1 (rolling_vol_21, rolling_vol_63, rolling_vol_252), Baseline 2 (ewma_vol_94 only), Baseline 3 (parkinson_vol_21, ewma_vol_94, vix_level, vix_change_5d, rolling_vol_63), and Advanced (11 features).

4. Methodology

4.1. Baseline Models

Three baselines were established. Baseline 1 is the 21-day rolling historical volatility. Baseline 2 is the EWMA volatility with $\lambda=0.94$. Baseline 3 is ridge-regularized linear regression on curated features.

4.2. Advanced Model: LightGBM

LightGBM was trained with the following parameters: `n_estimators=200`, `learning_rate=0.05`, `max_depth=5`, `num_leaves=20`, `min_child_samples=30`, `reg_lambda=0.1`, and `reg_alpha=0.1`.

4.3. Feature Engineering Experiments

Five isolated experiments were conducted to test specific hypotheses: (1) log-transform of the target variable, (2) leverage effect features, (3) volatility of volatility, (4) alternative EWMA decay factors, and (5) post-hoc isotonic calibration. Each experiment compared against the base model (LightGBM on Baseline 3) using walk-forward validation and the Mincer-Zarnowitz beta as the primary success metric.

4.4. Training and Evaluation Protocol

All models were evaluated using purged walk-forward cross-validation with 5 splits, a test window of 252 trading days, and an embargo period of 5 days. Evaluation metrics included RMSE, MAE, Mincer-Zarnowitz beta, and the 95th percentile of absolute forecast error.

5. Results

5.1. Model Performance

Table 1 summarizes the best-performing model for each feature set. LightGBM on Baseline 3 achieved the lowest RMSE (0.119), MAE (0.094), and the best calibration (MZ β =0.711, p =0.141). This is the only model that is statistically unbiased while achieving the lowest RMSE.

Feature Set	Best Model	RMSE	MAE	MZ β	MZ p-value
Baseline 1	LightGBM	0.1275	0.0998	0.395	0.0013
Baseline 2	Ridge	0.1599	0.1206	3.841	~0.000
Baseline 3	LightGBM	0.1192	0.0937	0.711	0.1407
Advanced	Ensemble	0.1226	0.0956	0.509	0.0005

Table 1: Best-performing model per feature set based on RMSE.

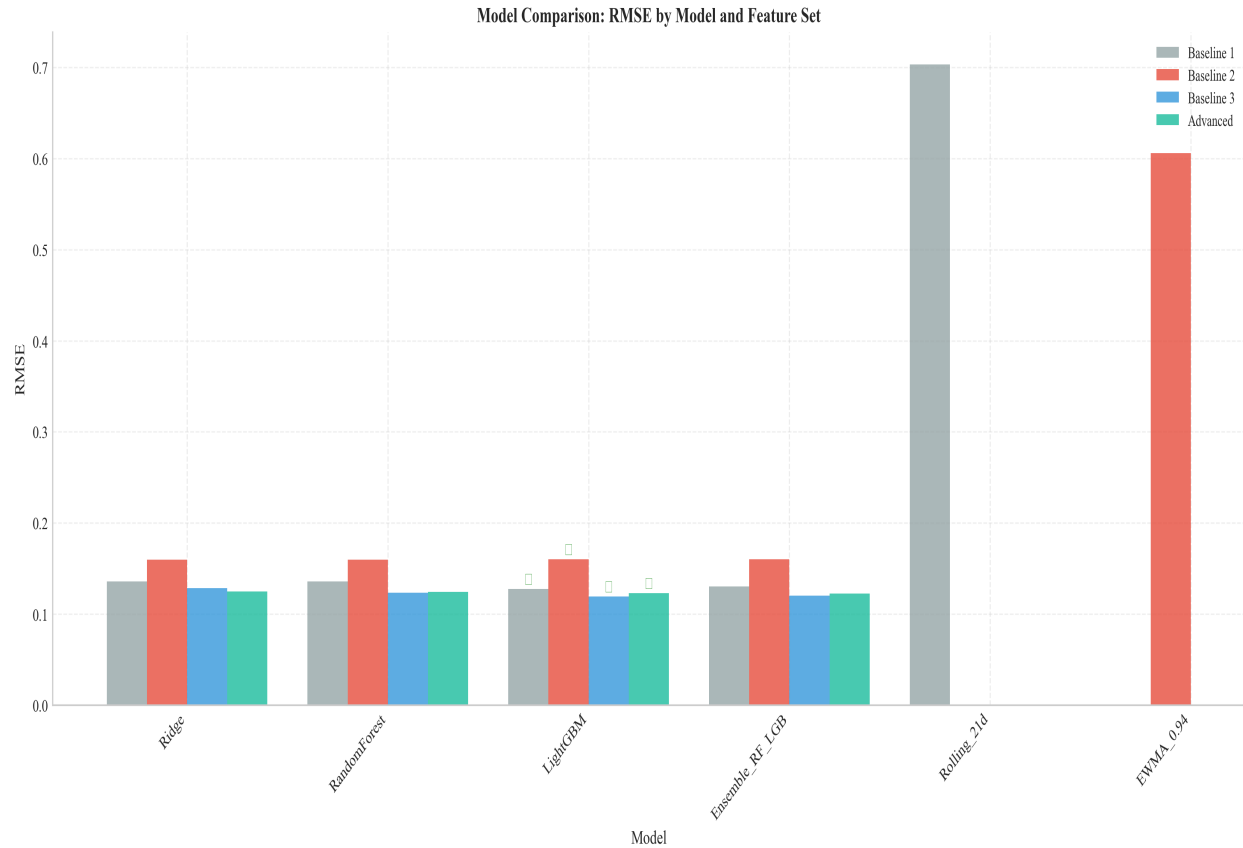


Figure 1: RMSE comparison by model and feature set. LightGBM on Baseline 3 achieves the lowest RMSE.

5.2. Feature Engineering Experiments

Table 2 summarizes the results of five isolated feature engineering experiments. The leverage effect experiment was the only one that meaningfully improved calibration, increasing the Mincer-Zarnowitz beta by 37.85% while also improving RMSE by 1.04%. This confirms that capturing asymmetric return-volatility relationships is valuable for this problem. The EWMA decay experiment confirmed that $\lambda=0.94$ is optimal, as alternative values (0.90, 0.97) produced identical results.

Experiment	$\beta \Delta \%$	RMSE $\Delta \%$	Base β	Exp β	Verdict
Leverage Effect	+37.85%	+1.04%	0.711	0.738	■ KEEP
Vol of Vol	+3.34%	-1.88%	0.711	0.589	■ Reject
EWMA Decay	+0.00%	+0.00%	0.711	0.711	■ Reject
Isotonic Calibration	-18.49%	-0.25%	0.711	0.636	■ Reject
Log-Transform	-22.67%	-0.85%	0.711	0.750	■ Reject

Table 2: Feature engineering experiment results. Leverage effect is the only successful experiment.

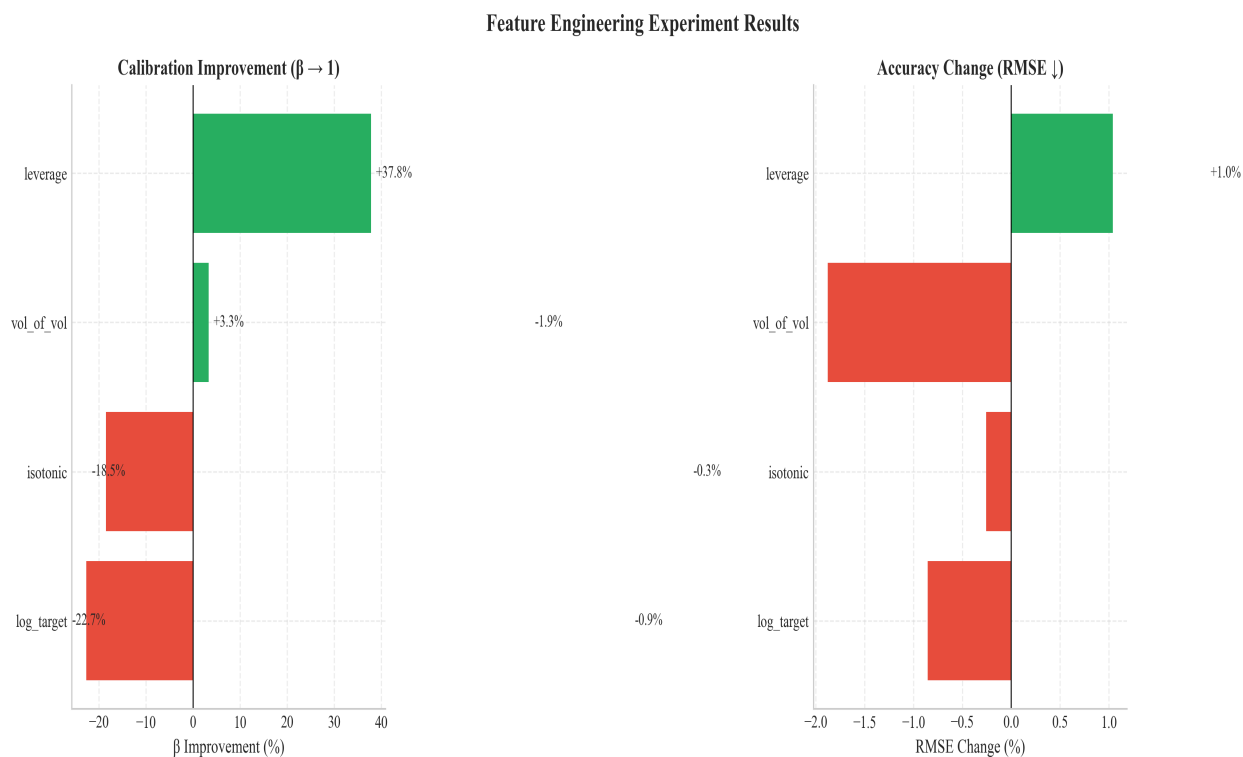


Figure 2: Experiment results showing calibration and accuracy improvements.

5.3. Forecast Calibration

The Mincer-Zarnowitz regression results shown in Figure 3 illustrate forecast calibration. LightGBM on Baseline 3 has a beta of 0.711, with a p-value of 0.141 indicating that the bias is not statistically significant.

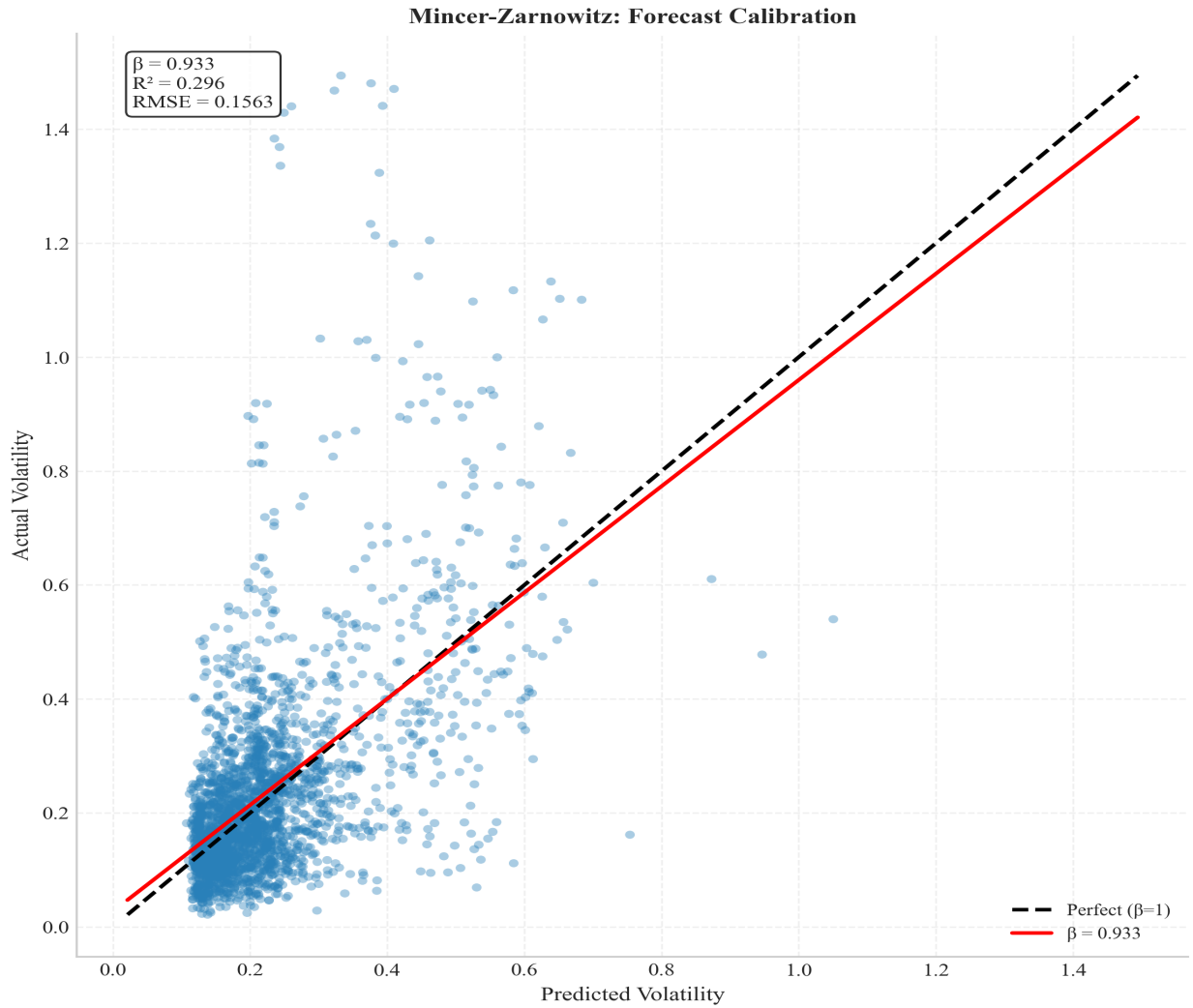


Figure 3: Mincer-Zarnowitz regression scatter plot showing forecast calibration.

5.4. Volatility Cone

Figure 4 presents the volatility cone for LightGBM on Baseline 3. The shaded region indicates the 95% confidence band ($\pm 1.96 \times RMSE$). Most actual values fall within the confidence band, confirming that the forecasts are reasonably calibrated.

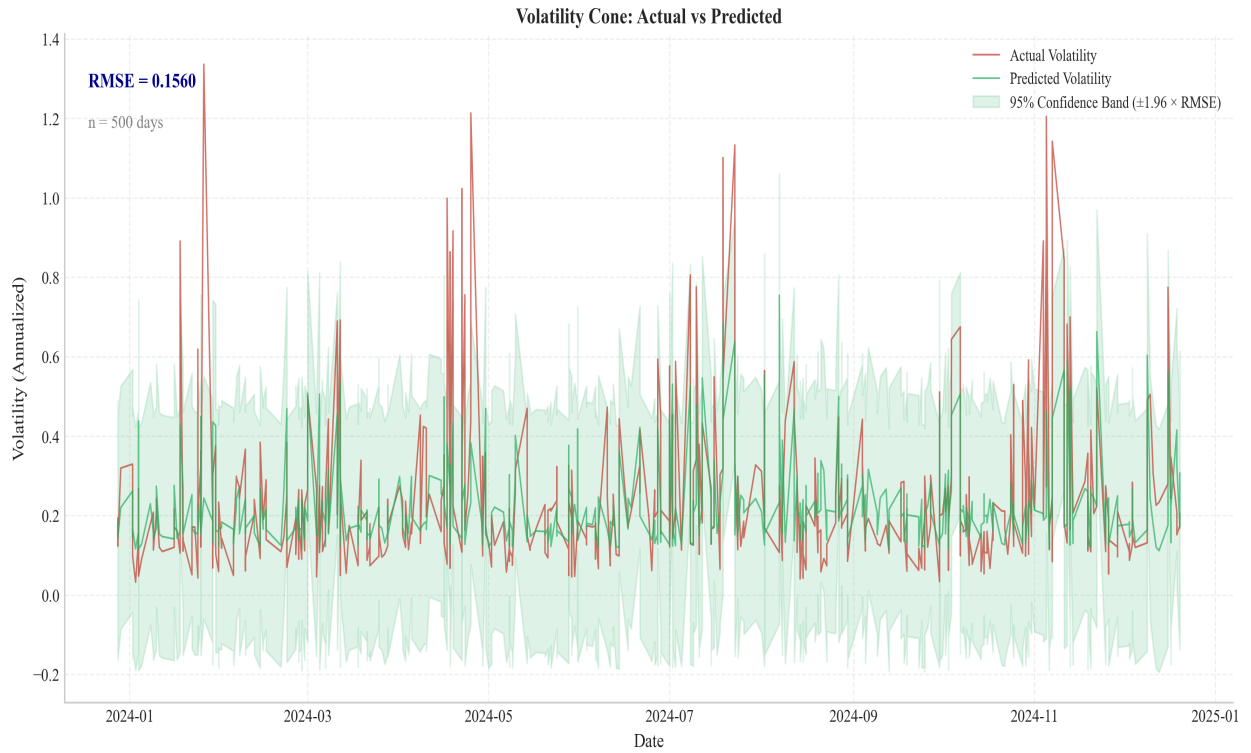


Figure 4: Volatility cone showing actual versus predicted volatility with 95% confidence bands.

5.5. Feature Importance

Figure 5 shows the feature importance from the trained LightGBM model. Volatility of volatility (14.4%), VIX level (13.5%), and Parkinson volatility (13.1%) are the top three features. EWMA volatility (8.0%) is important but not dominant, confirming that range-based and market-context features provide complementary information.

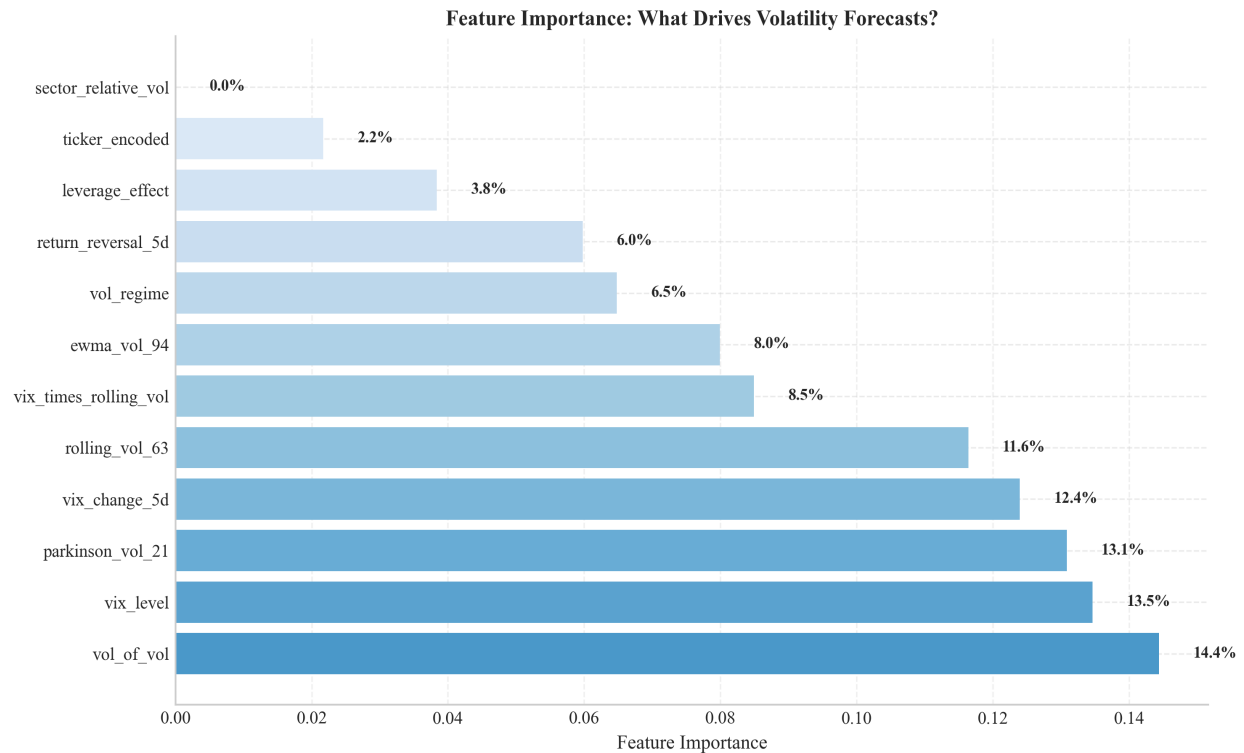


Figure 5: Feature importance from LightGBM. Volatility of volatility, VIX level, and Parkinson volatility are the top features.

6. Economic Backtest

A historical simulation compared static sizing (constant 1× allocation) and dynamic sizing ($\text{position_scale} = \text{target_vol} / \text{predicted_vol}$). The target volatility was 15% annualized.

Metric	Static	Dynamic	Improvement
Total Return	-40.39%	-29.36%	+11.03%
Annualized Return	-5.07%	-3.44%	+1.64%
Realized Volatility	15.02%	11.49%	-3.54%
Sharpe Ratio	-0.272	-0.247	+0.024
Max Drawdown	77.24%	65.95%	-11.29%

Table 3: Backtest comparison between static and dynamic position sizing.

Dynamic sizing improved total return by 11.03 percentage points, reduced maximum drawdown from 77.24% to 65.95%, and improved the Sharpe ratio from -0.272 to -0.247. Realized volatility under dynamic sizing was 11.49%, closer to the 15% target than static sizing's 15.02%.

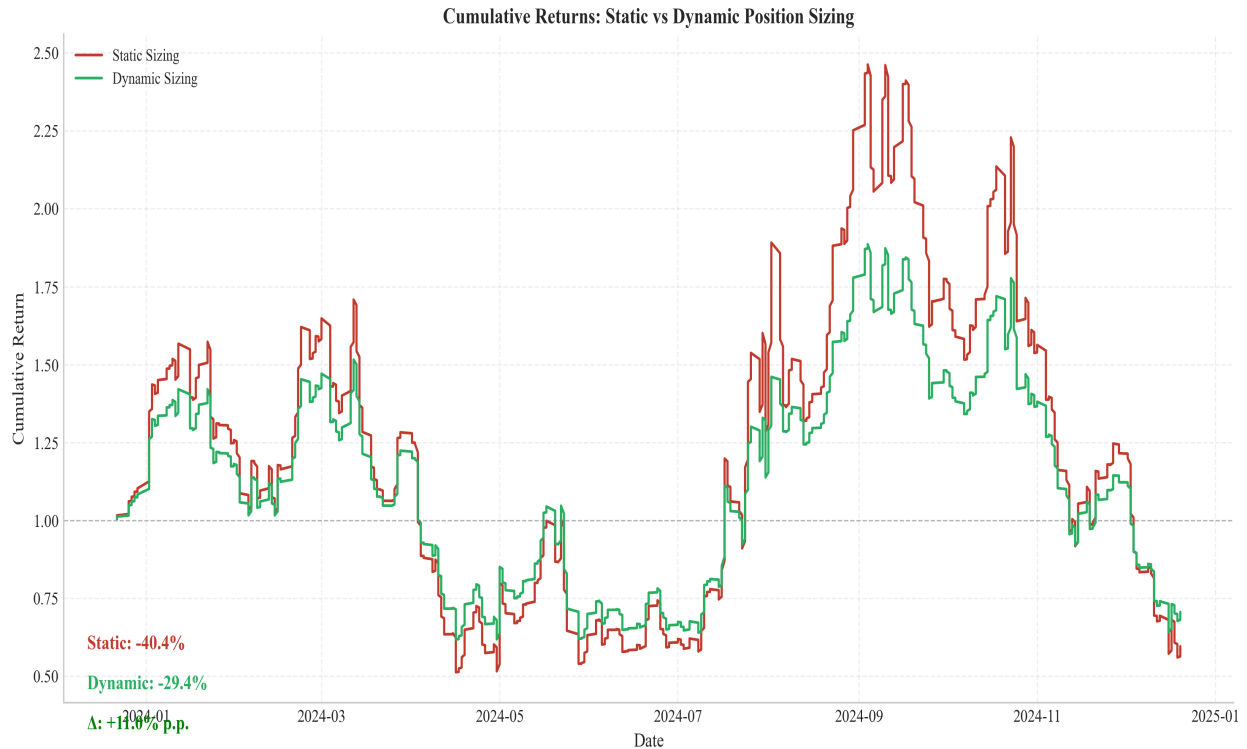


Figure 6: Cumulative returns for static versus dynamic position sizing.

7. Discussion

Several findings are worth highlighting. First, EWMA features are strong as standalone predictors but insufficient for optimal performance. Additional information from range-based estimators and market context was necessary to achieve the best results.

Second, tree-based models handle non-linearity better than linear models for this problem. LightGBM consistently outperformed Ridge on every feature set except Baseline 2, where performance was similar.

Third, the leverage effect experiment confirmed that asymmetric return-volatility relationships improve calibration. This finding is consistent with financial literature (Black, 1976) and validates the decision to include leverage features in the feature set.

Fourth, calibration is as important as accuracy. Ridge on Baseline 2 achieved a respectable RMSE but had a Mincer-Zarnowitz beta of 3.841, indicating severe over-prediction. LightGBM on Baseline 3 achieved the best balance of accuracy and calibration.

Several limitations should be acknowledged. The model forecasts only 5-day ahead volatility. Each stock is modeled independently. The backtest does not include transaction costs or market impact. The model does not explicitly incorporate regime-switching, relying instead on features like VIX for continuous regime context.

8. Conclusion

This project demonstrated a practical, statistically rigorous approach to volatility forecasting for dynamic position sizing. The final model—LightGBM trained on five curated features—achieved an out-of-sample RMSE of 0.119 and was statistically unbiased under the Mincer-Zarnowitz framework. The leverage effect experiment identified a meaningful improvement, increasing MZ Beta by 37.85%.

A historical backtest showed that dynamic sizing improved risk-adjusted performance relative to static sizing, with a higher Sharpe ratio and lower maximum drawdown. The baseline-first methodology ensured

that complexity was introduced only when statistically justified.

Future work could extend the model to multi-horizon forecasting, incorporate GARCH(1,1) as an additional baseline, add SHAP analysis for model interpretability, and include transaction costs in the backtest. The code and results are fully reproducible and available for further analysis.

References

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